VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-03 PHYSICS

Shailendra sir



Topics To Be Covered

1. REVISION
2. COMBINATION OF RESISTORS
3. HEAT ENERGY AND POWER
4. QUESTIONS


SHAILENDRA SIR
JOIN MY OFFICIAL TELEGRAM CHANNEL
JOIN NOW FOR PDF NOTES
@PHYSICSBY_SHAILENDRA_SIR
Physics Wallah



Revision

$$R_S = R_1 + R_2$$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$V = iR$$

QUESTIONS

The circuit diagram includes a 6 V battery, an ammeter A, a fixed resistor R_1 of 2Ω and a resistance wire R_2 connected between the terminals A and B. The resistance of the battery and ammeter may be neglected. Calculate the ammeter readings when the wire R_2 is of

(a) 0.20 m length and of resistance 4Ω.

(b) 0.40 m length and of same thickness and material as in case (a). $R \propto l$

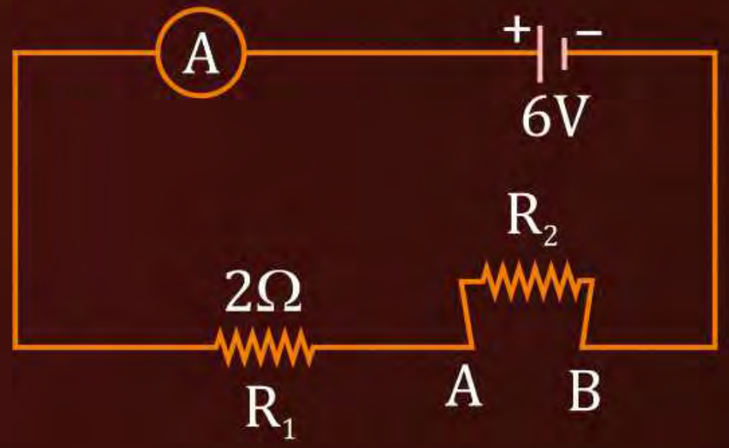
(c) 0.20 m length and of area of cross section double than in case (a).

$R \propto \frac{1}{A}$

(a) $I = \frac{V}{R} = \frac{6}{2+4} = 1\,A.$

(b) $I' = \frac{V}{R'} = \frac{6}{2+8} = 0.6\,A.$

(c) $I'' = \frac{V}{R''} = \frac{6}{2+2} = \frac{3}{2}\,A.$





QUESTIONS

In series combination of resistances:

1. p.d. is same across each resistance
2. total resistance is reduced
3. current is same in each resistance
4. all above are true.



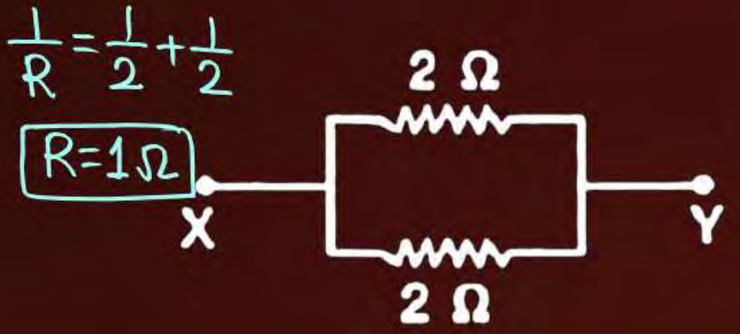
QUESTIONS

In parallel combination of resistances:

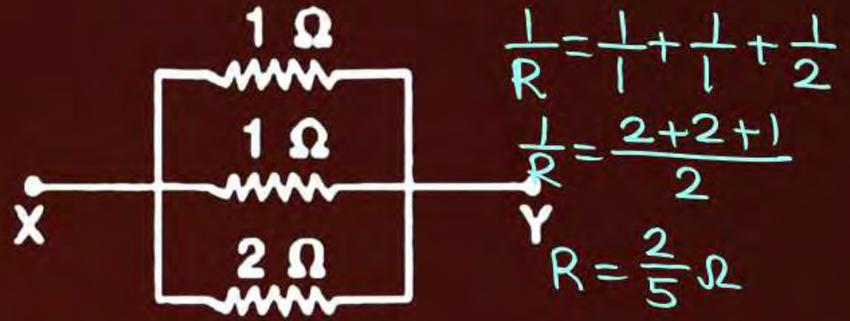
1. p.d. is same across each resistance
2. total resistance is increased
3. current is same in each resistance
4. all above are true.

QUESTIONS

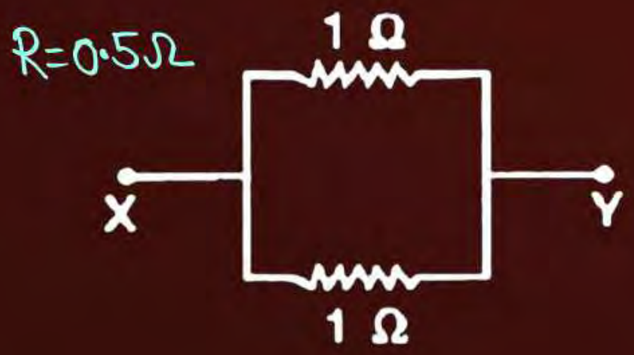
Which of the following combinations have the same equivalent resistance between X and Y?



(a)



(b)



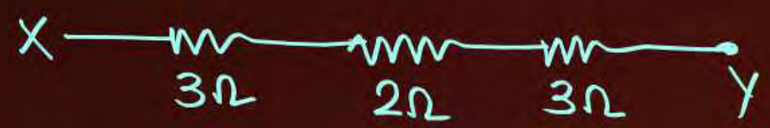
(c)



(d)

QUESTIONS

Five resistors, each of 3Ω, are connected as shown. Calculate the resistance (a) between the points P and Q, and (b) between the points X and Y.



$$R_{XY} = 3+2+3 = 8\Omega$$



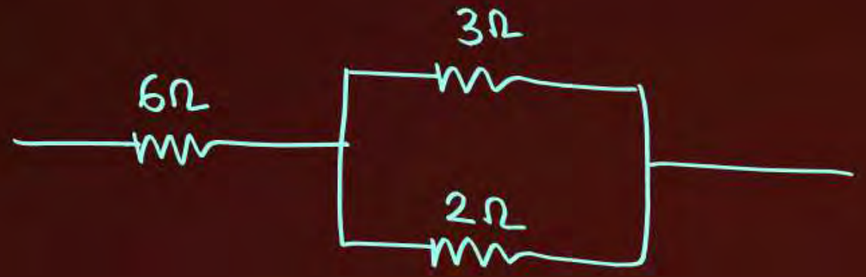
$$\frac{1}{R_{PQ}} = \frac{1}{6} + \frac{1}{3}$$

$$= \frac{1+2}{6}$$

$$R_{PQ} = 2\Omega$$

QUESTIONS

Three resistors of $6\Omega, 3\Omega$ and 2Ω are connected together so that the total resistance is greater than 6Ω, but less than 8Ω. Draw a diagram to show this arrangement and calculate its total resistance.



$$R_{eq} = 6 + \frac{2\times3}{2+3}$$

$$= 6 + \frac{6}{5}$$

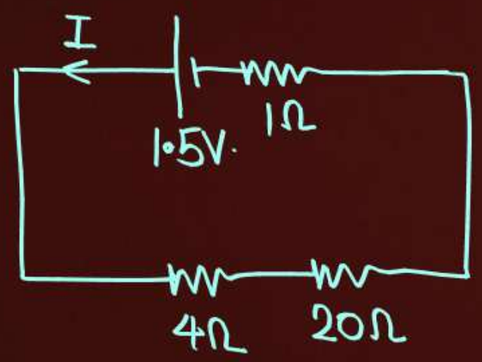
$$= \frac{36}{5}$$

$$= 7.2\Omega$$

QUESTIONS

A cell of e.m.f. 1.5 V ($\mathcal{E}$), internal resistance 1Ω (r) is connected to the resistors of 4Ω and 20Ω in series. Draw a circuit diagram and calculate:

(a) the current in the circuit,
(b) the p.d. across each resistor,
(c) the p.d. across the cell, and
(d) the voltage drop when the current is flowing.



(a) $I = \frac{V}{R} = \frac{1.5}{4+20+1} = \frac{1.5}{25} = 0.06\ A.$

(b) $V_{1\Omega} = iR = (0.06)(1) = 0.06\ V.$

$V_{20\Omega} = iR = (0.06)(20) = 1.2\ V.$

$V_{4\Omega} = iR = (0.06)(4) = 0.24\ V.$

(c) $V = \mathcal{E} - Ir$

$= 1.5 - 0.06 \times 1$

$= 1.44$ Volts.

(d) $v = ir$

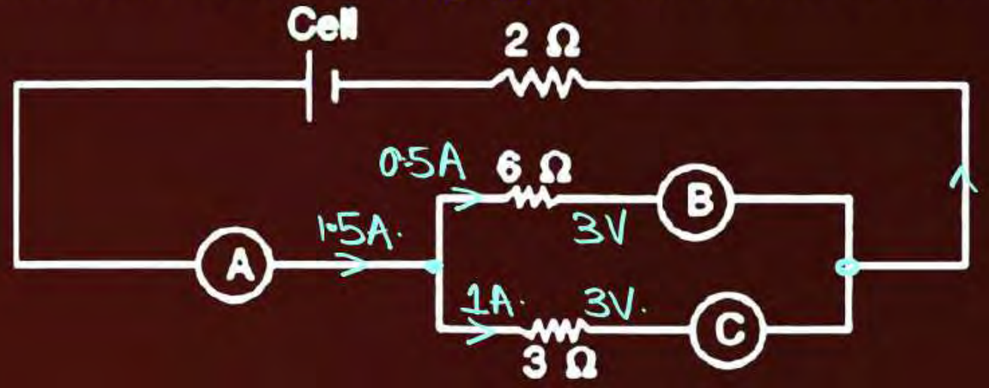
$= 0.06 \times 1$

$= 0.06\ V.$

QUESTIONS

In the diagram given below, A, B and C are three ammeters each of negligible resistance. The ammeter B reads 0.5 A . Calculate:
(a) the readings of the ammeters A and C,
(b) the total resistance of the circuit, and
(c) the e.m.f. of the cell.

1.5A → 1A.



$$R_{eq} = 2 + \frac{6 \times 3}{6+3} = 4\Omega$$

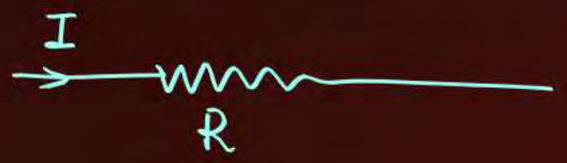
$$\mathcal{E} = i(R_{eq})$$

$$= 1.5 \times 4$$

$$= 6V.$$

$$V_{6\Omega} = iR = 0.5 \times 6 = 3V.$$

ELECTRICAL ENERGY



$V = \frac{W}{q}$

$W = qV$

$U = (It)V$

$\boxed{U = VIt}$

$U = (IR)(It)$

$\boxed{U = I^2Rt}$

$U = \left(\frac{V}{R}\right)^2 Rt$

$\boxed{U = \frac{V^2}{R}t}$

$q = It$

$\boxed{H = VIt = I^2Rt = \frac{V^2}{R}t}$

Unit : Joule or Cal.

$\boxed{1\,Cal = 4.18\,J.}$



ELECTRICAL POWER

$$P = \frac{W}{t} = \frac{U}{t}$$

$$P = VI = I^2R = \frac{V^2}{R}$$

$$P = \frac{VI\cancel{t}}{\cancel{t}} = VI$$

$$P = \frac{I^2R\cancel{t}}{\cancel{t}} = I^2R$$

$$P = \frac{\frac{V^2}{R}\cancel{*}}{\cancel{*}} = \frac{V^2}{R}$$

COMMERCIAL UNIT OF ELECTRICAL ENERGY

$$P = \frac{E}{t}$$

$$E = P \times t$$

P → kW, t → h

1 unit = 1 KWh = 3.6 MJ.

1 KWh = 1000 W × 1 h.

= 1000 W × (60 × 60) s.

$$1\ KWh = 3.6 \times 10^6\ J$$

POWER RATING OF ELECTRICAL APPLIANCES

Generally an electrical appliance such as electric bulb, geyser, heater, etc. is rated with its power and voltage. For example, an electric bulb is rated as **100 W–220 V**. **It means that if the bulb is lighted on a 220 V supply, the electric power consumed by it is 100 W** (100 J of electrical energy is converted in the filament of bulb into the light and heat energy in 1 second).

$I_{Bulb} = \frac{P}{V} = \frac{100}{220} = 0.45\ A.$

From this rating, we can calculate the following two quantities:

(1) the **resistance** of the filament of bulb when it is glowing, and

(2) the **safe limit of current** which can flow through the bulb while in use.

$P = VI$ → $\boxed{I = \frac{P}{V}}$

$P = \frac{V^2}{R}$ → $\boxed{R = \frac{V^2}{P}}$

$R_{Bulb} = \frac{(220)^2}{100} = 484\ \Omega$

QUESTIONS

An electric kettle is rated at 230V (V), 1000W (P). (a) What is the resistance of its element when in use? (b) What is the safe value of current that can pass through its element?

(a) $R = \frac{V^2}{P} = \frac{(230)^2}{1000} = 52.9\ \Omega$

(b) $I = \frac{P}{V} = \frac{1000}{230} = 4.35\ A$

QUESTIONS

V P

An electric iron is rated '220V, 1kW'. Under normal working conditions, find:

(i) the resistance of its heating element,

(ii) the amount of current that will flow through the element,

(iii) the amount of heat that will be produced in 5 minutes, and

(iv) the power consumed if the line voltage falls to 180 V.

$V \propto I$

imp ⇒

(i) $R = \frac{V^2}{P} = \frac{(220)^2}{1000} = 48.4\ \Omega$

(ii) $I = \frac{P}{V} = \frac{1000}{220} = 4.54\ A.$

(iii) $H = I^2 R t$

$= (4.54)^2 \times 48.4 \times (5 \times 60)$

$H = 2.99 \times 10^5\ J$

(iv) $P = VI$

$= 180 \times 4.54$

$P = \frac{V^2}{R}$

$= \frac{(180)^2}{48.4}$

$P = 669.4\ W.$



HEATING EFFECT OF CURRENT

The amount of heat produced in a wire on passing current through it, depends on

(1) **The amount of current passing through the wire:** The amount of heat H produced in a wire is directly proportional to the square of current I passing through the wire,

$$\mathbf{H \propto I^2}$$

(2) **The resistance of wire:** The amount of heat H produced in the wire is directly proportional to the resistance R of the wire, i.e.,

$$\mathbf{H \propto R}$$

(3) **The time for which current is passed in the wire** The amount of heat H produced in a wire is directly proportional to the time t for which current is passed in the wire i.e.,

$$\mathbf{H \propto t}$$

Combining the eqns.

$$\mathbf{H = I^2Rt}$$

where I is in ampere, R in ohm and t in second. The relation is also known as Joule's law of heating.

QUESTIONS

A voltage source sends a current 2.5 A to a resistor of 20Ω connected across it for 5 minutes. Calculate: (i) the p.d. of the source, (ii) the electrical energy supplied by the source, and (iii) the heat in cal, produced in the resistor.

(i) $\boxed{V = iR}$

$= 2.5 \times 20$

$= 50$ volts.

(ii) $\boxed{E = i^2Rt}$

$= (2.5)^2 \times 20 \times (5 \times 60)$

$= 37500$ J.

(iii) $E = \frac{37500}{4.18}$ cal.

$E = 8971.3$ cal.

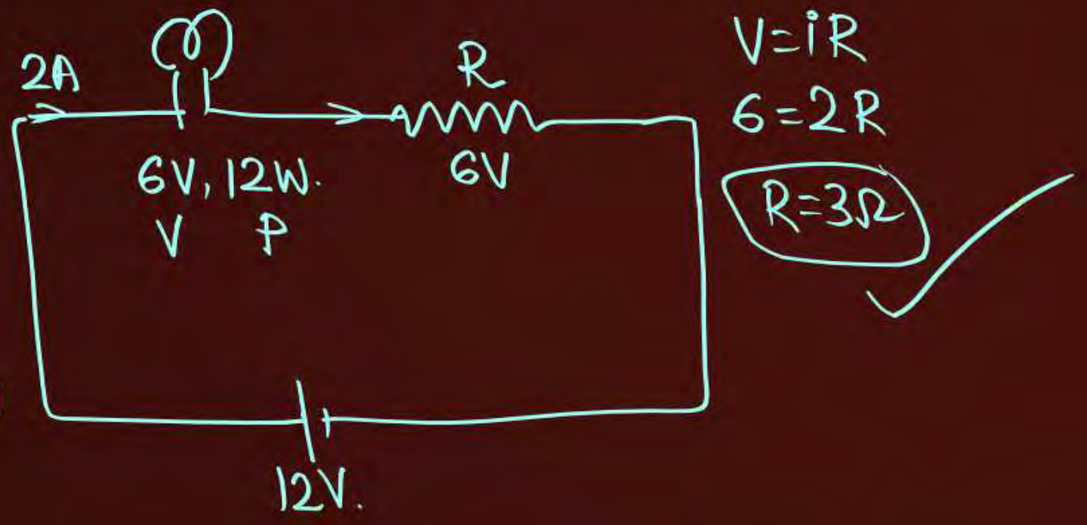
QUESTIONS

A 6V, 12W lamp is connected in series with a resistor R and a source of voltage 12 V.

(a) What is the purpose of the resistor R? R will consume extra 6V.

(b) Calculate the value of the resistor R, for the proper working of the lamp.

(c) What is the current flowing through the circuit?



$P = VI$

$12 = 6\ I$

$I = 2A$

$V = iR$

$6 = 2R$

$R = 3\Omega$

QUESTIONS

Two coils of resistances $R_1 = 3\Omega$ and $R_2 = 6\Omega$ are connected in series across a battery of p.d. 12 V. Find : (i) the electrical energy consumed in 1 minute in each resistance, and (ii) the total electrical energy supplied by the battery in 1 minute.

(i) $H_1 = I^2 R_1 t$

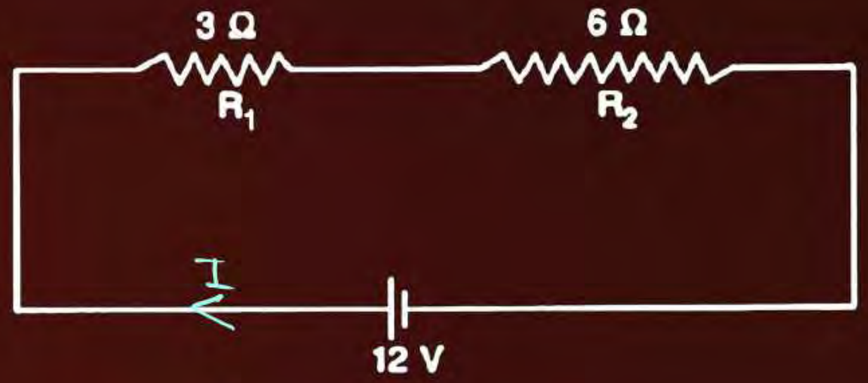
$= \left(\frac{4}{3}\right)^2 3 \times 60$

$= 320\ J.$

$H_2 = I^2 R_2 t$

$= \left(\frac{4}{3}\right)^2 (6)(60)$

$= 640\ J.$



$I = \frac{V}{R} = \frac{12}{3+6} = \frac{12}{9} = \frac{4}{3}\ A.$

(ii) $E = H_1 + H_2$

$= 320 + 640$

$= 960\ J.$

QUESTIONS

Two resistors R_1 and R_2 of resistance 3Ω and 6Ω respectively are connected in parallel across a battery of p.d. 12 V. Draw the circuit diagram. Calculate the electrical energy consumed in 1 minute in each resistance.

$$H_1 = \frac{V^2}{R_1} t$$

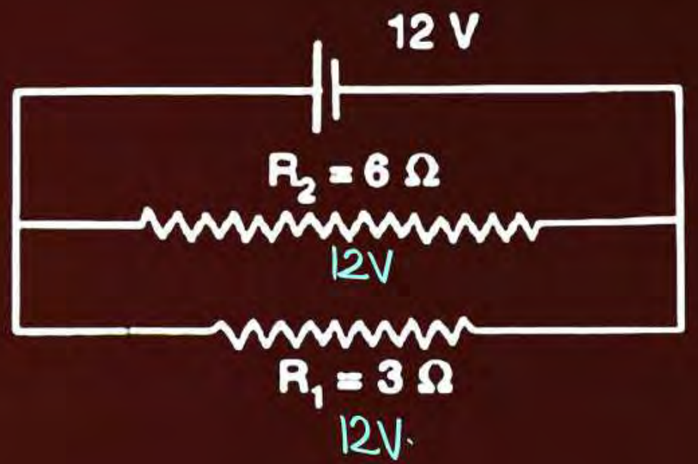
$$= \frac{12^2}{6} \times 60$$

$$= 1440 \text{ J}$$

$$H_2 = \frac{V^2}{R_2} t$$

$$= \frac{12^2}{3} \times 60$$

$$= 2880 \text{ J}.$$

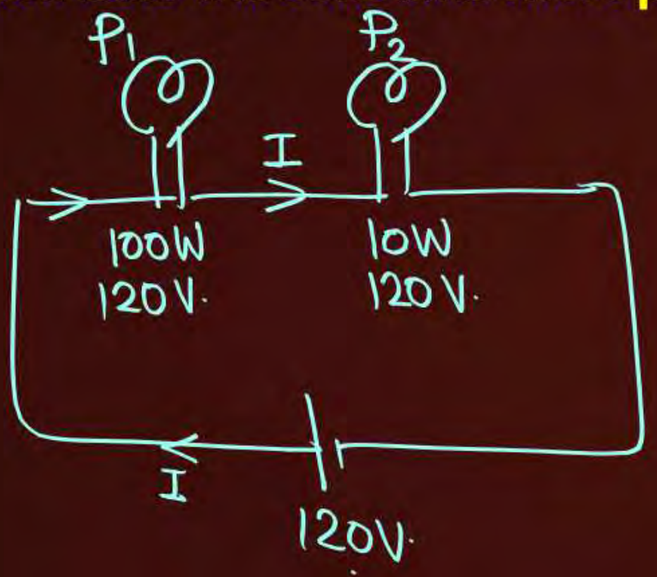


QUESTIONS

(Imp) Two bulbs A and B are rated 100W, 120V and 10W, 120V respectively. They are connected across a 120 V source in series.

(a) Calculate the current through each bulb.

(b) Which bulb will consume more power?



$$I = \frac{V}{R_1 + R_2}$$

$$= \frac{120}{144 + 1440}$$

$$I = 0.075\ A$$

$$P = I^2 R t$$

↓

same

$P \propto R$

$P_2 > P_1$

$$R_1 = \frac{V^2}{P} = \frac{120^2}{100} = 144\ \Omega$$

$$R_2 = \frac{120^2}{10} = 1440\ \Omega$$

QUESTIONS

$R_1 = 144\,\Omega$ $R_2 = 1440\,\Omega$

Imp Two bulbs A and B are rated $100\text{W}, 120\text{V}$ and $10\text{W}, 120\text{V}$ respectively. They are connected in parallel across a 120 V source.

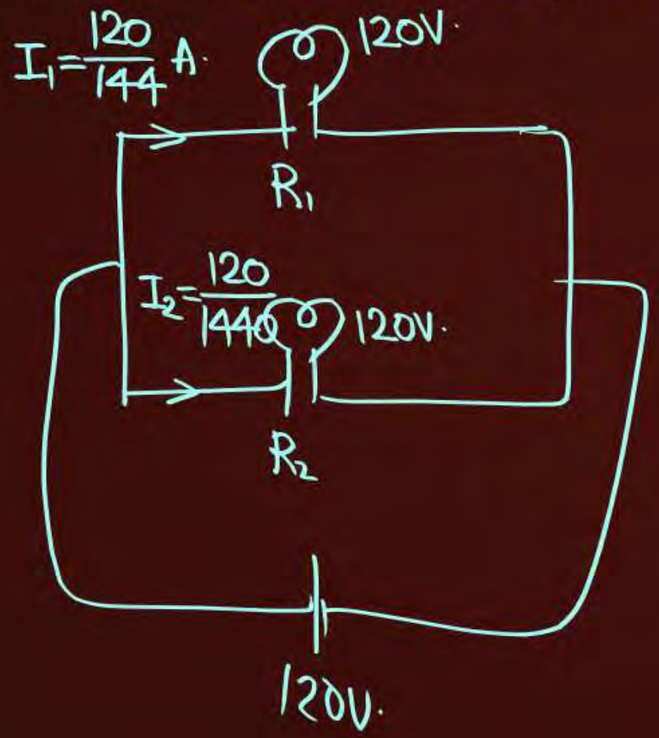
(a) Find the current in each bulb.

(b) Which bulb will consume more power?

$P = \frac{V^2}{R} t$ (same)

$P \propto \frac{1}{R}$

$P_1 > P_2$





QUESTIONS

A geyser has a rating $\underset{P}{2\text{kVA}}, \underset{V}{240\text{V}}$. (a) What is the electrical energy consumed by it in (i) kWh, and (ii) joule if it is used for 90 minutes. (b) If the cost of electricity is ₹4.50 per commercial unit, find the cost.

(a) $E = P \times t$

$= 2 \times \left(\frac{90}{60}\right)$

$E = 3\ \text{kWh}.$

$= 3 \times 3.6 \times 10^6$

$= 10.8 \times 10^6\ \text{J}.$

$E = 1.08 \times 10^7\ \text{J}$

(b) Cost $= 4.5 \times 3$

$= ₹\ 13.5$

QUESTIONS

(a) Calculate the electrical energy in kWh consumed in a month, in a house using 2 bulbs of 100 W each and 2 fans of 60 W each, if the bulbs and fans are used for an average of 10 hours each day.

(b) If the cost per unit is ₹ 4.50, calculate in part (a) the amount of electric bill to be paid per month.

(a) $E = P \times t$

$= \left(2 \times \frac{100}{1000}\right) \times (10 \times 30) + \left(2 \times \frac{60}{1000}\right)(10 \times 30)$

$= 60 + 36$

$= 96$ KWh.

(b) Amount $\Rightarrow 96 \times 4.5$

$= 432$ Rs

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You